

CJM410 WiFi Module AT Commands

Ver. 2.6

Revision History

Revision	Date	Description
1.0	May 2016	Initial release
1.1	June 2016	Add new commands for features
1.2	August 2016	Fix wording errors
1.3	August 2016	Support SoftAP mode and update commands
1.4	September 2016	Update command set and description format
1.5	May 2017	Add high speed UART port command.
1.6	July 2017	Update default values.
1.7	October 2017	Update AP/STA default security mode to WPA2
1.8	June 2018	Add debug UART port setting command
1.9	October 2018	Add new commands to get wireless interface MAC address and enable/disable debug UART port to print debug messages.
2.0	April 2020	Add new commands to emulate the action of WPS push button. Add new commands to support MQTT connection.
2.1	July 2020	Add new command to set max. STA connections in AP mode
2.2	June 2022	Add new commands to support HTTP client connection and methods.
2.3	June 2022	Add HTTPC_ADDHEADER=, HTTPC_CLEARHEADER=, and HTTPC_SETBODY= commands.
2.4	July 2022	Add SNTP client commands
2.5	December 2022	Add DNS server IP address command
2.6	April 2024	Add timeout value commands for network transmission and wireless status checking.

1. Command syntax format

<CMD>=[<para-1>][, <para-2>][, ...para-n]<CR><LF>

- CMD: Command
- Para-n: parameters.
- CR: return , '\r' or 0x0d.
- LF: new line, '\n' or 0x0a.
- < >: must
- []: option
- All characters are case sensitive.

2. Response message

+<RSP>[op][para-1, para-2,...para-n]<CR><LF>

- +: Response prefix
- RSP: the "ok" is successful, and the "err" is failed.
- op: '=' means the response has parameters.
- para-n: the result of query or error code and reason.
- CR: return , '\r' or 0x0d.
- LF: new line, '\n' or 0x0a.
- Error code:

Error code	Description
-1	Invalid command
-2	Invalid parameters
-3	Forbidden operation
-4	System operating error

3. AT Command Set

Command	Description
Mode switch	
+++	Enter command mode
ATO=	Enter data transparent mode
Wireless	
AT+WLINK=	Get wireless connection status in STA mode.
AT+WMAC=	Get wireless interface MAC address.
AT+WS=	Start WLAN scan. STA mode only
AT+WD=	Disassociate from connected AP. STA mode only
AT+WPS=	Emulate the action of WPS push button and start WPS connection
AT+WOP=?/<0/1/2>	Get/Set WLAN operation mode. (default: depend on firmware support)
	0 STA mode
	1 SoftAP mode
AT+WMODE=? / <b/g/n>	Get/Set WLAN PHY mode. (default: 802.11n mode)
AT+WAP=?/<ssid><channel>[,<hidden>]	Get/Set SoftAP mode settings.
	ssid Specify SSID between 1 ~ 32 characters. (default: CJN-WiFi)
	channel FCC: 1 ~ 11, CE: 1 ~ 13 (default: 6)
	hidden Hidden SSID
	0 Disable (default)
	1 Enable
AT+WAPSEC=?/<NONE/WEP/WPA/WPA2>	Get/Set security mode in SoftAP mode.
	NONE No Security
	WEP WEP mode
	WPA WPA version 1
	WPA2 WPA version 2 (default)
AT+WAPWEPKEY=?/<key index>,<hex key>	Get/Set WLAN WEP key in SoftAP mode
	key index Key index 1 ~ 4
	hex key Either 10 or 26 hexadecimal characters
AT+WAPWPA=?/<cipher>,<pass phrase>	Get/Set WPA/WPA2 settings in SoftAP mode
	cipher TKIP
	CCMP (default)
	passphrase (8 ~ 63) characters (default: 1234567890)

AT+WAPMAXSTA=?/<num>	Get/Set the max. number of stations allowed to connect	
	num	Between 1 ~ 5 (default: 5)
AT+WSTA=?/<ssid>[,<channel>]	Get /Set SSID for STA mode to associate	
	ssid	Specify SSID between 1 ~ 32 characters. (default: CJN-WiFi/"" for HTTPC)
	channel	Specified channel. (default: 0)
AT+WSTASEC=?/<NONE/WEP/WPA/WPA2>	Get/Set security mode in STA mode.	
	NONE	No Security
	WEP	WEP mode
	WPA	WPA version 1
	WPA2	WPA version 2 (default)
AT+WSTAWEPKEY=?/<key index>, <hex key>	Get/Set WLAN WEP key in STA mode	
	key index	The key index should be the same as AP specified default key index 1 ~ 4
	hex key	Either 10 or 26 hexadecimal characters
AT+WSTAWPA=?/<cipher>,<passphrase>	Get/Set WPA/WPA2 settings in STA mode	
	cipher	TKIP
		CCMP (default)
	passphrase	(8 ~ 63) characters (default: 1234567890)
Network		
AT+NTCPLINK=	Get the connection status of TCP client	
AT+NPING=<IP addr>	Ping to specified host	
	IP addr	IPv4 IP address
AT+NIPSET=? /<IP>,<Mask>,<GW>	Get/Set IP network parameters	
	?	Show current IP network parameters.
	IP	IP address (default: 192.168.1.10)
	Mask	Subnet mask (default: 255.255.255.0)
	GW	Gateway IP address (default: 192.168.1.10)
AT+NDHCPC=?/<0/1>	Get/Set DHCP client disabled or enabled. STA mode only. (default: 1)	
AT+NIPDNSS=?/<DNS1>[,<DNS2>]	Get/Set DNS server IP address.	
	DNS1	The primary DNS server IP address
	DNS2	The second DNS server IP address
AT+NSOCK=?/<TCP/UDP/MQTT/HTTPC>	Get/Set the network protocol. (default: depend on firmware support)	
AT+NTCPC=?/<IP>,<Port>	Get/Set the TCP client connection	

	IP	Remote server IP address
	Port	Remote server port number
AT+NTCPS=?/<Port>	Get/Set the TCP server connection	
	Port	Server port number. (default: 5000)
AT+NUDPC=?/<IP>,<Port>	Get/Set the UDP client connection	
	IP	Remote server IP address. (broadcast: 255.255.255.255)
	Port	Remote server port number
AT+NUDPS=?/<Port>	Get/Set the UDP server connection	
	Port	Server port number
AT+NTTO=<timeout>	timeout	Set the timeout value to wait for receiving data from UART port to transmit the received data to network. The value should be between 1 ~ 500 ms. (default: 100)
AT+NRCVTO=<timeout>	timeout	Set the timeout value to check the status of wireless connection when there is no data received from network. The value should be between 1 ~ 600 seconds. (default: 10)
AT+MQTTCONN=?/<server_url>,<port>,<cer_name>,<client_id>,<keepalive_interval>,<autoconnect_en>,<clean_session>,<username>,<passwd>]]]]]]]	Get/Set the parameters to connect to MQTT server	
	server_url	The URL on the server side, for example, 192.168.2.8/mqtt_base
	port	The port number on the server side.
	cer_name	When using the certification, set to the certification name; otherwise, set the key word of "ssl_disable" to disable it. (default: ssl_disable)
	client_id	The client ID name. (default: mqtt_XXXXXX) XXXXXX: the last 3 bytes of mac address
	Keepalive_interval	The keep-alive interval, in unit of seconds. (default: 10)
	autoconnect_en	Enable auto-connect or not. 0: disable; 1: enable. (default: 1)
	clean_session	Clean the last session environment or not. 0: disable; 1: enable. (default: 1)
	username	User name
	passwd	Password

AT+MQTTDISC=	Disconnect from the MQTT server	
AT+MQTTPUB=<topic>,<message>[,<message_id>[,<qos>[,<is_retained>[,<is_duplicated>]]]]	Publish the message about certain topic	
	topic	Topic string
	message	Message string
	message_id	Message ID
	qos	QoS level: 0/1/2
	is_retained	Message is retained or not. 0/1
	is_duplicated	Is duplicated message? 0/1
AT+MQTTSUB=<topic>,<qos>	Subscribe the topic	
	topic	Topic string
	qos	QoS level: 0/1/2
AT+MQTTUNSUB=<topic>	Unsubscribe the topic	
	topic	Topic string
AT+SSLC_START=	Start the SSL client, and return the SSL context index	
AT+HTTPC_CONN=<server>[,<ssl_ctx_index>[,<port>[,<timeout>]]]	HTTP client connects to HTTP server. If the connection is successful, it will return the client index number. At most, four clients can be created.	
	server	The server can be IPV4,IPV6 or domain name with https:// or http://
	ssl_ctx_index	The SSL context index. (default: 1)
	port	HTTP server listening port. (default: 80 for http, 443 for https)
	timeout	HTTP client request timeout value. Unit is millisecond. (default:8000)
	Ex, at+httpc_conn=https://httpbin.org,1	
AT+HTTPC_DISC=<client_num>	Disconnect from HTTP server	
	client_num	Index of created client
	Ex, at+httpc_disc=1	
AT+HTTPC_GET=<client_num>,<url>	HTTP client GET method	
	client_num	Index of created client
	url	The URL to access
	Ex, at+httpc_get=1,/index.html	
AT+HTTPC_POST=<client_num>,<url>	HTTP client POST method	
	client_num	Index of created client
	url	The URL to access

	Ex, at+httpc_post=1,/index.cgi	
AT+HTTPC_PUT=<client_num>,<url>	HTTP client PUT method	
	client_num	Index of created client
	url	The URL to access
	Ex, at+httpc_put=1,/index.cgi	
AT+HTTPC_PATCH=<client_num>,<url>	HTTP client PATCH method	
	client_num	Index of created client
	url	The URL to access
	Ex, at+httpc_patch=1,/index.cgi	
AT+HTTPC_SETPARAM=<client_num>,<key>,<value>	HTTP client set parameter. Before posting, the key value pair can be updated using this command.	
	client_num	Index of created client
	key	The name of key to be set
	value	the value to be set to key
	Ex, at+httpc_setparam=1,ssid,myWiFi	
AT+HTTPC_ADDHEADER=<client_num>,<hdr_name>,<hdr_value>	HTTP client add header for request methods	
	client_num	Index of created client
	hdr_name	The name of header to be set
	hdr_value	The value to be set to header
	Ex, at+httpc_addheader=1,Content-Type,application/json	
AT+HTTPC_CLEARHEADER=<client_num>	HTTP client clear header	
	client_num	Index of created client
	Ex, at+httpc_clearheader=1	
AT+HTTPC_SETBODY=<client_num>,<content>	HTTP client set body	
	client_num	Index of created client
	content	The content of body
	Ex, at+httpc_setbody=1,{"success": "true"}	
AT+SNTPC=?/<0/1>	Get/Set SNTP client to be disabled or enabled. (default: HTTPC firmware is 1, others is 0)	
	0	disable
	1	enable
	Default public SNTP server addresses are time.google.com and time.cloudflare.com	
AT+SNTP_ZONE=?/<UTC String>,<dse>	Get/Set the time zone and enable/disable daylight saving. (default: UTC+08:00,0)	
	UTC String	UTC time format should be UTC+XX:XX or UTC-XX:XX

	dse	Daylight saving enable or disable. 0: disable 1: enable		
AT+SNTP_GETTIME=	Get the current timestamp from the SNTP server.			
AT+SNTP_GETTIMEOFDAY=	Display the time in seconds.			
UART port				
AT+UART=?/<baud rate>	Get/Set debug UART port baud rate			
	Baud rate	Supported baud rate: 9600/19200/38400/57600/115200 (default: 115200)		
AT+HSUART=? / <baud rate>[,<flow control>]	Get/Set high speed UART port baud rate and RTS/CTS flow control			
	baud rate	Supported baud rate: 115200/230400/380400/460800/921600/ 1843200/2000000/3000000 (default: 115200)		
	Flow control	Enable or disable RTS/CTS flow control		
		0	Disable (default)	
		1	Enable	
AT+DBG=?/<0/1>	Get/Set debug UART port to print debug messages enabled or disabled. (default: 0)			
Management				
AT+ECHO=	Echo all the input to console.			
AT+RST=	Reset the system			
AT+SAVE=	Save all the settings to flash			
AT+DEF=	Restore all the settings to factory default and reboot the system			
AT+VER=	Get firmware version			
AT+APPLY=	Save all settings to flash and reboot the system. This command is equal to (AT+SVAE) + (AT+RST) commands			
AT+OTA=<IP>,<File Name>	Firmware upgrade via wireless by TFTP protocol			
	IP	Remote TFTP server IP address		
	File Name	File name of firmware located at TFTP server working directory		

Note:

1. If the operation mode is set to STA mode, after applying the settings, if the

module can't connect to the AP successfully in 10 seconds, the system will restore the module to AP mode automatically. This function is disabled in HTTPC firmware support.

2. The MQTTCONN setting is only supported in MQTT enabled firmware.
3. The HTTP client connection and methods are only supported in HTTPC enabled firmware.

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